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Mortality prediction via age-specific band selection

Yechao Meng[a], Liqun Diao[b] and Chengguo Weng[b]

[a]School of Mathematical and Computational Sciences, University of Prince Edward Island, Charlottetown, Prince Edward Island, Canada; [b]Department of Statistics and Actuarial Science, University of Waterloo, Waterloo, Ontario, Canada

ABSTRACT

A novel mortality prediction framework, age-specific band selection, is proposed to borrow information from 'neighboring' ages and train prediction models tailored for each individual age in a mortality table. This framework is further extended to borrow information across multiple populations through two proposed approaches: a distance-based approach and an ACF model-based approach. Extensive empirical studies with the Human Mortality Database are conducted to illustrate the enhanced prediction accuracy achieved by these methods.

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Mortality prediction; human mortality database; age-specific band selection; clustering; K-nearest neighbor

1. Introduction

Actuaries face a significant challenge in obtaining accurate future mortality rate prediction for insurance product design, pricing, and other relevant risk management purposes. While the past decades have witnessed a proliferation of various mortality models, there is still a demand for improved prediction performance. This paper aims to contribute in this direction by establishing novel data-driven procedures that select pertinent mortality data for borrowing information used in extrapolative models so as to improve their accuracy of forecasting.

Prominent among the available models is the Lee-Carter mortality projection model (R. D. Lee & Carter, 1992) and its extensions (Booth et al., 2002; Jong & De Tickle, 2006; R. Lee & Miller, 2001; Wilmoth, 1993). For senior age groups (e.g. 65 or larger), the Cairns-Blake-Dowd (CBD) model (Cairns et al., 2006) and its generalizations (e.g.Cairns et al., 2009) are widely recognized for their superior predictive performance. These models can be considered as vehicles for borrowing information across ages, and their estimation involves the specification of age ranges for mortality data. Recent literature underscores the impact of different age range choices on predictive results, such as the contrast between full-age range and partial-age range models conducted by Shang and Haberman (2020).

An analysis of mortality data from the Human Mortality Database reveals notable similarities in development patterns within specific age groups (e.g. young, middle, and senior ages) and distinct differences across these groups. Existing mortality models demonstrate that capturing age-specific patterns can improve prediction accuracy. For example, multi-factor models incorporate additional age-period terms (Booth et al., 2002; Hyndman & Ullah, 2007; Renshaw & Haberman, 2003) to better capture distinct mortality development features across different age groups. Another class of methods, the spline-based approaches, models human mortality non-parametrically within the framework of generalized additive models. Local mortality patterns and trends are captured through localized

CONTACT Chengguo Weng c2weng@uwaterloo.ca Department of Statistics and Actuarial Science, University of Waterloo, M3-200 University Avenue West, Waterloo, Ontario N2L 3G1, Canada

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basis functions, while a penalty term enforces smoothness in the resulting mortality surface. Currie et al. (2004), Camarda (2012), Dodd et al. (2018), Camarda (2019), and Zhu and Zhou (2023). More recent efforts have explored alternative approaches to capturing similarities in mortality development parttern by focusing on filtering the mortality data and emphasizing the most relevant information. For instance, Tsai and Cheng (2021) incorporated statistical clustering methods into mortality models to enhance predictive performance. Their approach involves performing clustering analysis to group ages with similar characteristics and then applying the Lee-Carter model to each age cluster independently. Similarly, Wang et al. (2021) considered the 'neighboring' effect, operating under the assumption that the future mortality trajectory of a target age is highly correlated with the development patterns of its neighboring age.

This paper contributes to the strand of research that filters mortality data and introduces a novel age-specific band (ASB) selection framework for mortality forecasting. It breaks down the overall prediction objective into separate tasks for individual ages, and in each separate task, it focuses on the prediction of one individual age that we call the target age. Based on the assumption that mortality development patterns are more similar at adjacent ages than at ages farther apart, it is desirable for a target age to borrow information from neighboring ages to enhance prediction accuracy. The ASB framework considers borrowing information from age bands centered at the target age of different spans, and the chosen age band is expected to embrace, to the largest extent, the potential benefits of the similarity in mortality development patterns shared by the target age and the ages within the band. Additionally, a smoothing procedure is incorporated into the ASB framework to mitigate disparities in the band sizes learned for consecutive target ages.

While our ASB method, like the clustering methods and P-spline approaches, is capable of borrowing information from neighboring ages and detecting local structure to enhance mortality modeling, it differs in several aspects. In the ASB method, a distinct set of ages is selected specifically for each target age to improve mortality forecast accuracy. In contrast, clustering methods aim to group ages that are more homogeneous within each group (or cluster) than across groups. The P-spline method was originally developed to smooth mortality rates over age and time without imposing any specific model structure. It achieves a parsimonious description of mortality development trends using basis function expansions, with global smoothness enforced through a penalty on the coefficients. A 2-dimensional P-spline framework can capture localized patters across both age and time dimensions. In contrast, our ASB framework focuses solely on local structures across age, and does so by explicitly selecting reference ages for each target age to borrow information, which makes the information-borrowing process transparent and targeted. Another key distinction between the ASB and the P-spline methods lies in the prediction procedure. The ASB method aligns with traditional mortality modeling framework based on time series projection, whereas the P-spline method are based on extrapolating the basis functions, where the shape of the basis functions on the boundary can significantly influence future mortality projections.

The proposed ASB-based mortality prediction framework is adaptable for integrating some existing mortality forecasting models. As an illustrative example, we utilize the Lee-Carter model as the embedded model in this study. To assess the performance of the ASB-based prediction framework, we analyze mortality data from 24 populations of both genders in the Human Mortality Database (HMD), alongside benchmarks that utilize statistical learning methods such as clustering. Empirical results confirm the effectiveness of the proposed ASB approach, as it consistently improves predictive accuracy across the majority of the 48 populations studied, as measured by test errors.

Leveraging information across populations has gained prominence and proven beneficial when applied with appropriate methods, as demonstrated by prior studies (Diao et al., 2021, 2023; Hyndman et al., 2013; Kleinow, 2015; J. Li, 2013; J. S.-H. Li & Hardy, 2011; N. Li & Lee, 2005; Russolillo et al., 2011; Shang & Hyndman, 2017; Zhou et al., 2014), among others. An additional significant contribution of this paper is to explore potential extensions of the ASB method to multi-population settings for information-borrowing across multiple populations and ages simultaneously. We introduce two specific approaches for implementation: a distance-based method, and an augmented

common factor (ACF) model-based approach that integrates the age band framework with geographic data. Extensive empirical comparisons using the Human Mortality Database confirm that both extended methods improve predictive accuracy compared to the ASB method applied to single populations. Notably, the distance-based approach yields greater accuracy improvements for forecasting female populations, while the ACF model based approach performs better for male populations.

The remainder of this paper is organized as follows. Section 2 provides a detailed description of the proposed ASB-based mortality prediction framework, along with the data-mining procedures used to determine the optimal age band selection. Section 3 presents empirical studies of the proposed methodologies, comparing them against benchmark models based on analysis of the Human Mortality Database. Section 4 introduces two specific approaches designed to borrow information in a multi-population context, accompanied by empirical studies. Section 5 concludes the paper, and the Appendix presents a step-by-step description of a hypothesis test used in empirical studies.

2. Age-specific band method for single-population data

In mortality table prediction, conventional approaches typically involve estimating a mortality model using data from the entire age range and then applying the estimated model to generate predictions for all ages. While this method is widely used, it may not fully capture the variations in mortality development patterns across different age groups, particularly when using single-factor models such as the Lee-Carter model. Although multi-factor models can better account for these age-specific patterns, the search for effective alternative models remains ongoing. Rather than increasing model complexity to address these variations, we propose the Age-Specific Band (ASB) framework as an alternative. This framework focuses on selecting the most relevant data ranges for each age, thus maintaining the model's simplicity while enhancing its predictive accuracy. The key innovation of this method lies in its ability to provide age-specific solutions without overcomplicating the mortality model.

Within the ASB framework, the overall prediction task is divided into separate tasks for each age in the mortality table. For each target age, dedicated machine learning algorithms are used to identify an optimal age band, from which relevant information is drawn to enhance mortality prediction accuracy specifically for that age.

2.1. General age-specific set selection

Before delving into the ASB-based mortality prediction framework, we first describe a broader framework that selects a set of ages, not necessarily an age band, for the mortality prediction of each individual age. This concept is essential for eventually extending the ASB approach from single-population context to a multiple-population setting. Let $\mathcal{X} := \{x_L, x_L + 1, \ldots, x_U\}$, where $0 \le x_L < x_U \le 100$, represents the age range in the mortality table that we aim to predict. Within this framework, mortality prediction is conducted separately for each individual age $x_0 \in \mathcal{X}$. For each target age x_0, we select an age-specific set $\mathcal{A}^*_{x_0}$ which includes the target age x_0, to be used in the estimation of a mortality model through a procedure we will describe later. Assume that mortality data in the study lies in an age-year window $\mathcal{X} \times \mathcal{T}$ with $\mathcal{T} = \{t_L, t_L + 1, \ldots, t_U\}$. For target age $x_0 \in \mathcal{X}$, a mortality model is estimated only using the data over the age-year window $\mathcal{A}^*_{x_0} \times \mathcal{T}$. The proposed framework is versatile and can be embedded with many single-population mortality forecasting models. When the Lee-Carter model or the CBD model is embedded, we do the estimation regarding the following formulation and data range:

$$\log m(x,t) = a_x + b_x k_t + \epsilon_{x,t}, \quad x \in \mathcal{A}^*_{x_0}, \quad t \in \mathcal{T}, \tag{1}$$

or

$$\operatorname{logit} q(x,t) = \log\left[\frac{q(x,t)}{1-q(x,t)}\right] = K_t + (x - \bar{x})k_t + \epsilon_{x,t}, \quad x \in \mathcal{A}^*_{x_0},\ t \in \mathcal{T}$$

where $m(x,t)$ denotes the central death rate, also known as the age-specific death rate, for age $x \in \mathcal{X}$ and year $t \in \mathcal{T}$ of a population, $q(x,t)$ is the probability that an individual of a population aged $x \in \mathcal{X}$ will die between t and $t+1$ given the individual is alive at time $t \in \mathcal{T}$, and $\epsilon_{x,t}$'s are independent and identically distributed residuals with zero mean and finite age-specific variance σ_x^2. The mortality prediction for the target age x_0 is then obtained by extrapolating the estimated mortality model. The prediction for the mortality table is formed by repeating the procedure of age set selection, model estimation, and extrapolation for each individual age. In the reminder of the paper, we focus on the Lee-Carter model as the embedded model within the proposed age-specific set selection framework, given its prominence in the literature for mortality prediction.

2.2. Age-specific band method

One approach to defining $\mathcal{A}_{x_0}^*$ for the target age x_0 is to leverage information from neighboring ages. The rationale behind this approach stems from the observation that adjacent ages typically exhibit more similarity in mortality development patterns compared to ages that are further from the target age x_0. This insight leads us to define $\mathcal{A}_{x_0}^*$ as an age band that includes the target age x_0 and its neighboring ages.

2.2.1. Symmetric age band

A convenient choice for the age band is a symmetric one, centered at the target age x_0, as defined below:

Definition 2.1 (Symmetric Age Band): For a given target age x_0, a symmetric age band of x_0 with a band radius r (where r is a nonnegative integer), denoted as $\mathcal{A}_{x_0}(r)$, consists of ages from $\tilde{x}_L$ to $\tilde{x}_U$, where

$$\tilde{x}_L = \max(x_0 - r, x_L) \quad \text{and} \quad \tilde{x}_U = \min(x_0 + r, x_U).$$

Note that the bound ages $\tilde{x}_L$ and $\tilde{x}_U$ in the above definition are specific to x_0. For simplicity, we suppress the dependence of these notations on the target age x_0 in the subsequent discussion, unless otherwise stated.

As the name suggested, a symmetric age band includes an equal number of ages from both sides of the target age, unless the target age is nearby the smallest or largest age in the mortality data range. For a given target age x_0, different band radii r lead to distinct age bands $\mathcal{A}_{x_0}(r)$, reflecting different extents of information-borrowing from neighboring ages. Therefore, the task of determining a desirable symmetric age band for each target age x_0 can be viewed as identifying the age-specific value of the hyperparameter r, denoted as $r_{x_0}^*$. It is well-known that the mortality development patterns between two adjacent ages are highly similar. As a result, we do not expect the value of $r_{x_0}^*$ to vary significantly when x_0 changes within a small neighborhood.

Indeed an asymmetric age band of x_0, with potentially different values of the lower radius r_L and the upper radius r_U, can be considered for further extensions. One possible approach to determine an optimal asymmetric age band involves using a fully data-driven Deletion-Substitution-Addition (DSA) algorithm, which provides a computationally efficient solution. This DSA algorithm is conceptually similar to the one presented in Diao et al. (2021), which was developed for the population selection problem in multi-population mortality modeling. However, to maintain our primary focus, we will limit our anlysis to symmetric age bands.

2.2.2. Selection of age band by block cross-validation

As discussed previously, age band selection is essentially a procedure of determining the value of the hyperparameter r for symmetric age bands. A natural approach for selecting r is to use a cross-validation procedure with a temporal modeling-validation data split. In this approach, the available time series of mortality data is partitioned into two chronological segments. The first segment is

used to build mortality models with different values of the hyperparameter r, while the second segment is reserved for assessing the prediction performance of each model. The optimal value of r is then selected as the one that achieves the best prediction performance when validated on the second segment of data.

One limitation of this temporal modeling-validation split paradigm is that it provides only a single estimate of model performance, which can be less reliable due to potential overfitting or underfitting to the chosen validation set. This issue may also explain why multiple-fold cross-validation is commonly used, even in cross-sectional data where the entries of data are independently distributed and carry no temporal dependence.

To address the limitations of the temporal modeling-validation split, we adopt block cross-validation (BCV), which offers several key advantages for time series modeling (Bergmeir & Benítez, 2012; Bergmeir et al., 2014) and has become one of main tools in this field. The method has also recently gained attention in the mortality prediction literature, where it has been used as a robust tool for validating prediction error (Kessy et al., 2022; SriDaran et al., 2022). By partitioning the training data into multiple modeling and validating blocks and excluding data points surrounding each validating block, BCV mitigates the risk of information leakage caused by temporal autocorrelation in time series data. This approach also accommodates non-stationarity in the data by validating model performance across distinct time intervals, and thus, provides a better assessment of model prediction power under varying temporal conditions. Additionally, by averaging performance metrics over multiple validating blocks, BCV reduces the bias and variance inherent in single-split validation, offering a more robust estimate of model accuracy. These strengths make BCV particularly well-suited for our mortality modeling task.

The BCV procedure we adopt has two hyperparameters u and s, where u denotes the number of blocks into which the procedure divide the training data set, and s denotes the size of the data dropped during the procedure. To perform the BCV procedure, the available training mortality data is sequentially partitioned into u blocks of roughly equal size. Each block is used as a validation set to evaluate the prediction performance of models trained on the remaining blocks, which we call modeling set. The overall prediction performance of the models is determined by aggregating the prediction results across all u validation blocks. In line with the literature (Bergmeir & Benítez, 2012; Bergmeir et al., 2014), we incorporate data dropping into the BCV procedure. With a pre-selected value of s, s-year data surrounding the validation block is excluded from the modeling set. The primary purpose of the data dropping is to reduce the influence of short-term temporal dependencies between the modeling and validation sets.

The selection of u involves a trade-off between bias and variance, as in the conventional multiple fold cross validation. Fewer blocks result in larger modeling sets, reducing bias but increasing variance due to fewer validation instances. Conversely, more blocks reduce variance by providing more validation sets but can increase bias due to smaller modeling sets. Similarly, the choice of s balances two objectives: (1) reducing short-term dependence between the modeling and validation data sets, and (2) preserving as many data points as possible for modeling. Excluding more data points reduces short-term dependence but may lead to smaller modeling sets and thus increase bias. Retaining more data improves modeling but may allow temporal correlations and thus lead to overoptimistic performance estimates.

Figure 1 illustrates the partitioning of data into modeling and validation blocks for a 4-fold BCV with 5 years of data dropping, as implemented in the empirical study described in Section 3. The empirical study involves training annual mortality data spanning 41 years, from 1970 to 2010. In the BCV procedure, the dataset is divided into four non-overlapping blocks: 1970–1979, 1980–1989, 1990–1999, and 2000–2010. Each of these blocks, represented by the blue squares in Figure 1, serves as a validation block in turn to assess the prediction performance of models with different age bands. For each validation block, data within 5 years ($s = 5$) on either side of the block in the remaining three blocks is excluded, as indicated by the green circles in Figure 1. The remaining data after this exclusion, shown as orange triangles in Figure 1, is used to train the prediction model. The optimal

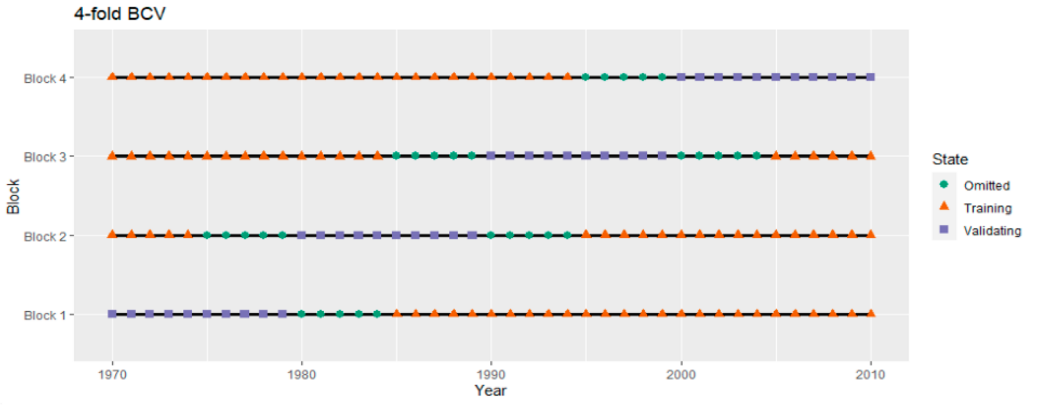


Figure 1. Illustrative figure of a 4-fold block cross-validation with $s = 5$. Orange triangles represent data points from the time series used for model training, blue squares stand for data points used for validation, and green circles represent the omitted data points.

age band for a target age is then selected based on the collective performance across all four validation blocks, evaluated using the BBCV-SSE measure defined in Equation (4) below.

Using mortality data in the modeling blocks, we calibrate the Lee-Carter model (1) by applying a standard SVD procedure to obtain the fitted values of age-specific coefficients a_x and b_x and the period index k_t, and fit a random walk with drift

$$k_t = k_{t-1} + d + \epsilon_t,$$

where d serves as the drift parameter and ϵ_t as the noise term. Alternative approaches to calibrating the coefficients a_x, b_x and k_t include likelihood-based methods that minimize a loss function based on the number of deaths. However, as noted by Santolino (2023), a useful rule of thumb is that the choice of calibration method should align with the scale of the objective variable. This supports our decision to adopt the SVD-based approach, which minimizes the squared error in the approximated logarithmic mortality rate matrix, as our primary objective is to predict future logarithmic mortality rates.

Next, we project k_t onto the validation block. If the validation block is not the first block in time, the forward-fill approach is employed. The projection is anchored on the period index k_{t_0}, where t_0 represents the last year in the modeling period preceding the validation block. For example, $t_0 = 1974$ when Block 2 is the validation block, and $t_0 = 1984$ when Block 3 is the validation block, as shown in Figure 1. The projected period index, $\hat{k}_t$, for calendar year t in the validation block is computed by

$$\hat{k}_t = k_{t_0} + (t - t_0)\hat{d}, \tag{2}$$

where $\hat{d}$ is the fitted value of the drift parameter, and $t - t_0$ reflects the time difference between the predicted time point t in the validation block and the nearest time point t_0 in the preceding modeling set.

If the validation block is the first block in temporal order, the backward fill approach is employed. In this case, the projected k_t is given by

$$\hat{k}_t = k_{t_1} - (t_1 - t)\hat{d}, \tag{3}$$

where t_1 is the first year in the modeling set following the validation block. For instance, $t_1 = 1985$ when Block 1 in Figure 1 serves as the validation block.

Let $\mathcal{T}^{(v)}_{validate}$ denote the set of years spanned by the v-th validation block, and $\mathcal{T}^{(v)}_{modelling}$ represent the set of years included in the corresponding modeling data set. The block cross-validation sum of

squared error (BCV-SSE) for a given age band $\mathcal{A}_{x_0}(r)$ is calculated as

$$\text{BCV-SSE}(\mathcal{A}_{x_0}(r)) = \sum_{v=1}^{u} \sum_{t \in \mathcal{T}_{validate}^{(v)}} \left[\log m_{x_0,t} - \log \hat{m}_{x_0,t}^{(v,r)}\right]^2, \tag{4}$$

where $m_{x_0,t}$ denotes the observed central death rate for age x_0 and year t, and $\hat{m}_{x_0,t}^{(v,r)}$ is the corresponding predicted value obtained from the mortality model trained using data from $\mathcal{A}_{x_0}(r) \times \mathcal{T}_{modelling}^{(v)}$. It is important to note that the predicted value $\hat{m}_{x_0,t}^{(v,r)}$ depends on the method used to project the k_t sequences during the BCV procedure, and consequently, so does the BCV-SSE measure. Additionally, the resulting BCV-SSE value is inherently influenced by specific partitioning of the training data into u folds. In our empirical study presented in later sections, we consistently follow the partitioning scheme illustrated in Figure 1. Furthermore, while we use the sum of squared error (SSE) as the performance metric in Equation (4), other metrics such as mean absolute error (MAE), mean squared percentage error (MSPE), and mean absolute percentage error (MAPE) could also be considered. There is no specific actuarial reason for choosing SSE in our study. We selected it as an example primarily because it is one of the most commonly used loss metrics for measuring the discrepancy between statistical predictions and observed data.

The optimal age band $\mathcal{A}_{x_0}^*$ is chosen as $\mathcal{A}_{x_0}(r_{x_0}^*)$, where the band radius $r_{x_0}^*$ minimizes BCV-SSE:

$$r_{x_0}^* = \text{argmin}_r \, \text{BCV-SSE}(\mathcal{A}_{x_0}(r)).$$

The model fitting and cross-validation procedure, described for the Lee-Carter model above, can also be adapted to other mortality models, such as the CBD model.

2.2.3. Smoothing

Implementing the BCV procedure for different target ages generates a sequence of optimal radii $r_{x_0}^*$. Due to the inherent randomness of the realized BCV-SSE values, the sequence $r_{x_0}^*$ can exhibit variability and lead to roughness and bumpiness as the target age x_0 varies. As we have argued previously, it is reasonable to expect the optimal age bands of adjoining target ages to be similar, and thus it is reasonable to require some smoothness in the $r_{x_0}^*$ sequence.

To address the smoothness expectation, we apply a smoothing procedure to the sequence $r_{x_0}^*$ via a 'moving average' scheme. In this approach, we replace each radius with $r_{x_0.\textbf{smooth}}^*$, which is the integer rounded from the average of some adjacent points in the original sequence:

$$r_{x_0.\textbf{smooth}}^* = \frac{1}{2h+1}\left(r_{x_0-h}^* + r_{x_0-h+1}^* + \cdots + r_{x_0+h}^*\right)$$

where h is radius of the smoothing window. For target ages nearby the boundaries x_L or x_U, the averaging is applied to a truncated section of the sequence, either from $r_{x_L}^*$ to $r_{x_0+h}^*$, or from $r_{x_0-h}^*$ to $r_{x_U}^*$, as appropriate.

Different values of the hyperparameter h will result in varying degrees of smoothness. The choice of h can be also determined using a BCV procedure. In the empirical study in Section 3, we will evaluate the impact of different smoothing window sizes and implement the BCV procedure for selecting the optimal value of h.

3. Empirical study with age-specific band method

In this section, we empirically assess the performance of the proposed ASB method using the Human Mortality Database. We conduct a comprehensive comparison of the ASB method with the Lee-Carter model and the clustering approaches introduced in Tsai and Cheng (2021).

3.1. Empirical setting and predictive models

We consider the mortality data from 24 populations listed in Table 1, spanning the age range $\mathcal{X} = \{0, 1, \ldots, 100\}$ and the time period $\mathcal{T} = \{1970, 1971, \ldots, 2019\}$. This time frame is selected to ensure the use of the recent data while avoiding potential biases from structural changes in mortality patterns before 1970 (e.g.R. Lee & Miller, 2001; Tuljapurkar et al., 2000) and the significant mortality upsurge associated with the COVID-19 pandemic, of which the long-term effects have not yet been fully studied. The geographic information presented in the table will not be used in this section, but it will play a role Section 4.3 in a multi-population setting.

For our empirical study, the dataset is divided into a training set covering the years 1970 to 2010, and a test set encompassing the years 2011 to 2019. Each prediction model is trained using the training dataset (1970–2010) and then extrapolated into the testing period (2011–2019) to generate mortality predictions.

The prediction models used for comparison in this empirical study include:

- **Lee-Carter Model**: As a benchmark, it is fitted separately for each population and gender using the full-age band from 0 to 100.
- **LC-AgeCluster**: This benchmark model adapts the Lee-Carter approach by dividing the population into three distinct age clusters – Teen (ages 0–25), Adult (ages 26–85), and Elder (ages 86–100) –and fitting the model separately for each cluster.
- **ASB Models**: The ASB method is implemented with the Lee-Carter model embedded to perform mortality predictions for each population. The ASB method determines optimal age bands for each target age using a BCV procedure. Two variations are considered:
  - **LC-ageband**: It applies a 4-fold BCV procedure to identify the optimal age band for each target age. No smoothing is applied to the resulting band radius sequence $r^*_{x_0}$.
  - **LC-ageband-smooth**: A moving average smoothing procedure with a smoothing window radius $h = 3$ is applied to the band radius sequence $r^*_{x_0}$. As detailed in Subsection 3.3 later, the prediction performance remains quite stable in response to the change in the smoothing window size. The value of $h = 3$ is selected based on a uniform level of smoothness across all 24 populations and treating h as a hyperparameter determined through a data-driven BCV procedure. Specifically, h is chosen to minimize the overall BCV-SSE across all populations. In Subsection 3.3, we will compare the prediction results of the ASB method using a uniform smoothing window radius h across all populations with those obtained using population-specific h values determined through BCV in the smoothing procedure. The results indicate that population-specific h values do not lead to improved forecasting accuracy.
- **Clustering-based models**: These models cluster ages into groups based on their mortality dynamics and apply the Lee-Carter model to each cluster separately. The clustering process

Table 1. 24 populations from HMD and their geographic information.

Target Population	Geographic Group	Target Population	Geographic Group
Australia	Oceania	Netherlands	West Europe
Austria	West Europe	New Zealand	Oceania
Japan	Asia	Norway	Scandinavia
Belgium	West Europe	Poland	East Europe
Scotland	Great Britain	Portugal	South Europe
Canada	North America	U.S.A.	North America
Czech Republic	East Europe	Slovakia	East Europe
Denmark	Scandinavia	Spain	South Europe
Finland	Scandinavia	Sweden	Scandinavia
France	West Europe	Switzerland	West Europe
Hungary	East Europe	Taiwan	Asia
Italy	South Europe	England & Wales	Great Britain

uses annual increments in log mortality rates:

$$Y_{x,t} = \log m(x,t) - \log m(x,t-1), \quad t \in \{t_L+1, t_L+2, \ldots, t_U\}. \tag{5}$$

For a given age x, the sequence $Y_{x,t}$ is assumed as an independent sample from a normal distribution, with location and variability characterized by mean μ_x and variance σ_x^2, respectively. When the pairs $(\mu_{x_1}, \sigma_{x_1}^2)$ and $(\mu_{x_2}, \sigma_{x_2}^2)$ display similarity, the two ages x_1 and x_2 are deemed suitable to be put within the same cluster. In light of this, the clustering algorithms are executed on a set of two-dimensional objects $\{(\hat{\mu}_x, \hat{\sigma}_x^2) : x \in \mathcal{X}\}$, where $\hat{\mu}_x$ is the estimated mean and $\hat{\sigma}_x^2$ is the estimated variance, respectively calculated as:

$$\hat{\mu}_x = \frac{1}{t_U - t_L} \sum_{t=t_L+1}^{t_U} Y_{x,t}, \tag{6}$$

$$\hat{\sigma}_x^2 = \frac{1}{t_U - t_L - 1} \sum_{t=t_L+1}^{t_U} \left(Y_{x,t} - \hat{\mu}_x\right)^2. \tag{7}$$

Depending on the clustering method employed, three distinct prediction models are utilized. The model labels and their corresponding clustering methods are described below:

- **LC-Kmeans**: The K-means clustering method is used to form age clusters, with the number of clusters selected based on the recommendation from the `NbClust` function in the `R` package `NbClust` (Charrad et al., 2014).
- **LC-HCluster**: The Ward's agglomerative clustering method (Ward Jr, 1963) is applied to establish age clusters. This method employs a bottom-up merging approach using the sum of squared errors as the objective function on the two-dimensional objects $(\hat{\mu}_x, \hat{\sigma}_x^2)$. The number of clusters is also determined by the recommendation from the `NbClust` function in the `R` package `NbClust` (Charrad et al., 2014).
- **LC-GMM**: The Gaussian mixture model (GMM) clustering method is utilized to determine age clusters. The GMM approach assumes a mixture of Gaussian distributions for the data. Through iterative adjustments of the parameters of the mixture distribution to best align with the underlying data, each component distribution of the mixture models represents a distinct cluster. The GMM clustering algorithm comprises three key elements: (1) Initialization through clustering results from the agglomerative hierarchical clustering method; (2) Maximum likelihood estimation using the EM algorithm; (3) Model selection and determination of the number of clusters via approximate Bayes factors with the BIC. We adhere to the strategy introduced by Fraley and Raftery (2002) for fitting data with a Gaussian mixture structure, employing the `mclust` function within the `R` package `mclust` (Scrucca et al., 2016). Further details about this method can be found in Fraley and Raftery (2002) and Tsai and Cheng (2021).

3.2. Predicting performance

For a given population, we let $e(t)$ denote its age-aggregated test SSE and caculate it by

$$e(t) = \sum_{x \in \mathcal{X}} \left[\log m(x,t) - \log \widehat{m}(x,t)\right]^2, \quad t = 2011, \ldots, 2019, \tag{8}$$

where $m(x,t)$ represents the actual mortality rate and $\widehat{m}(x,t)$ denotes its predicted counterpart. Here $\mathcal{X} = \{0, 1, \ldots, 100\}$ is the full range of ages. The overall prediction accuracy is assessed through overall test SSE, which is computed as $\sum_{t=2011}^{2019} e(t)$. This measures the mortality prediction accuracy for one population. For a comprehensive comparison of prediction accuracy among different models, we

Table 2. Summary statistics of test SSEs of 24 female populations and 24 male populations comparing the prediction performance of different predicting methods.

	1st Quartile	Median	Mean	3rd Quartile
	Female Population			
Lee-Carter	25.44	59.35	60.87	90.55
Lee-Carter-AgeCluster	25.44	60.02	60.05	87.02
LC-GMM	24.72	64.10	61.18	91.90
LC-Kmeans	25.03	58.97	60.51	91.36
LC-HCluster	25.52	59.15	60.69	91.73
LC-ageband	**22.79**	61.14	59.79	86.51
LC-ageband-smooth	23.92	**58.37**	**57.71**	**84.61**
	Male Population			
Lee-Carter	31.03	60.98	59.06	77.01
Lee-Carter-AgeCluster	31.85	58.31	55.05	71.74
LC-GMM	24.59	52.13	49.11	**66.62**
LC-Kmeans	29.89	**51.18**	53.28	71.48
LC-HCluster	27.09	55.42	53.43	72.24
LC-ageband	**20.39**	55.76	49.60	71.87
LC-ageband-smooth	21.38	52.62	**48.14**	72.13

Notes: Bold numbers highlight the minimum values in each column.

compute the overall test SSEs for all 24 populations and report the 1st quartile, median, sample mean, and 3rd quartile of the resulting 24 SSEs in Table 2. Smaller values of these statistics in this context imply better prediction performance.

Table 2 reveals that the proposed ASB method with smoothing (LC-ageband-smooth) exhibits advantages in prediction accuracy compared to other methods, evidenced by its smallest mean of the resulting test SSEs in both female and male populations. The clustering-based methods also show improvement compared to the Lee-Carter model but in a smaller extent compared to the proposed ASB approaches, i.e. LC-ageband and LC-ageband-smooth. Furthermore, the LC-AgeCluster model improves upon the original Lee-Carter model by separately applying the Lee-Carter model to each of the three age groups (Teen, Adult, Elder). While it performs comparably well with the clustering methods, it is inferior to the ASB methods in terms of prediction accuracy.

Regarding the clustering methods, while Tsai and Cheng (2021) found in their empirical studies on mortality data in the US and the UK for both genders that the LC-GMM method is the best among the three clustering methods, our results do not fully support this conclusion. In our empirical studies, considering the overall performance over all the 24 populations, the LC-GMM method performs the best for male populations but not for the female populations, as judged by the mean of the 24 SSEs).

Additionally, we employ the one-sided Diebold-Mariano (DM) test introduced by Diebold and Mariano (1995) and Harvey et al. (1997) to compare the predictive accuracy of two forecasting models. A step-by-step description for the general procedure of the DM test is given in Appendix.

We divide the six predictive models described in Section 3.1 into two groups: group one consists of the two ASB methods, and group two consists of the benchmark models, namely the Lee-Carter model, the Lee-Carter model applied to each of the three age groups, and the three clustering models. We compare the prediction accuracy of a pair of models, one from each group, by conducting two one-sided DM tests based on age-aggregated test SSEs from two comparing models. One test assesses the null hypothesis that the model from the first group is no better than the one from the second group and the other test evaluates the null hypothesis that the model from the second group is no better than the one from the first group. If the former null hypothesis is rejected, we conclude that the model from group one is better than the one from group two and thus the model from group one wins the comparison; if the latter null hypothesis is rejected, the model from group two wins; if none of the two hypotheses are rejected, no model wins the comparison. For each pair of models, winner determination is repeated for all 24 populations and a higher frequency of winning indicates a superior prediction performance. In Table 3, we report the comparison results between two groups of models, the ASB models listed in the row and the benchmark models listed in the column. In each

Table 3. Number of wins for comparisons between the proposed models (in the row) and the benchmark models (in the column) based on pairs of one-sided DM tests.

	Lee-Carter	LC-GMM	LC-Kmeans	LC-HCluster	LC-AgeCluster
			Female Population		
LC-ageband	(11, 4)	(8, 6)	(9, 4)	(9, 4)	(10, 6)
LC-ageband-smooth	(14, 0)	(12, 1)	(15, 2)	(17, 1)	(17, 2)
			Male Population		
LC-ageband	(15, 4)	(7, 6)	(12, 6)	(13, 5)	(12, 2)
LC-ageband-smooth	(19, 3)	(11, 7)	(16, 4)	(16, 3)	(18, 2)

cell of Table 3, we report two integers, of which the first is the number of wins by the model in the row, and the second is the number of wins by the model in the column. out of 24 populations. Table 3 shows that the ASB models notably outperform the benchmark models in delivering more accurate predictions evidenced by their dominating numbers of wins. This advantage is particularly evident for the ASB model with a smoothing procedure.

We are also interested in examining how the performance of the proposed ASB methods varies across age groups. To the end, we divide the entire age spectrum into six non-overlapping age groups as follows:

- **Child:** ages 0 to 9;
- **Teenage:** ages 10 to 19;
- **Young Adult:** ages 20 to 29;
- **Adult:** ages 30 to 59;
- **Retiree:** ages 60 to 89;
- **Elder:** ages larger than 89.

We compute the age-aggregated test SSEs defined in (8) over a refined age group, i.e. with $\mathcal{X}$ in the formula (8) taken as one of the above age groups, and calculate the overall test SSE accordingly. We depict boxplots of overall test SSEs (logarithmically scaled) of 24 populations in Figures 2 and 3. Subfigures in the left panels correspond to female populations and the ones on the right correspond to male populations. Subfigures in Figure 2 provide results for Child, Teenage, and Young Adult age groups and those in Figure 3 for Adult, Retiree and Elder age groups. In these graphs, a boxplot leaning toward a leftward position indicates greater prediction accuracy. As depicted in Figures 2 and 3, the degree of improvement in predicting accuracy by using the proposed ASB methods varies across genders and age groups. Obvious improvements in prediction accuracy are observed for both female and male populations in the 'Adult' and 'Retiree' age groups, covering ages 30 to 89. The improvement appears more pronounced for male populations. However, prediction performance remains comparable to other methods in other age groups. This suggests that the proposed ASB method can enhance predictions for midage groups while delivering predictions of at least equal quality for other target ages compared to the benchmarks.

The previous model comparison analysis indicates that, while the ASB models achieve better age-aggregated SSEs over the 24 populations and yield an obvious outperformance over benchmarks, particularly for the age groups 'Adult' and 'Retiree', they generate comparable forecast accuracy for most other age groups. To illustrate the outperforming cases, we focus on both the male and female populations of Canada and England & Wales as examples. We compare the performance of the ASB model ('LC-ageband-smooth') with the Lee-Carter model. Figures 4 and 5 present the fitted and forecasted logarithmic mortality rates for ages 0, 70, and 80, using both models. As previously mentioned, the training set includes mortality data from 1970 to 2010, while the test period spans 2011 to 2019. The figures demonstrate that the ABS method provides closer alignment with the realized mortality values during the training period, and that the forecasts from the ABS method are also generally more accurate, or at least comparable, to those of the Lee-Carter model across all age-population combinations shown in the figures.

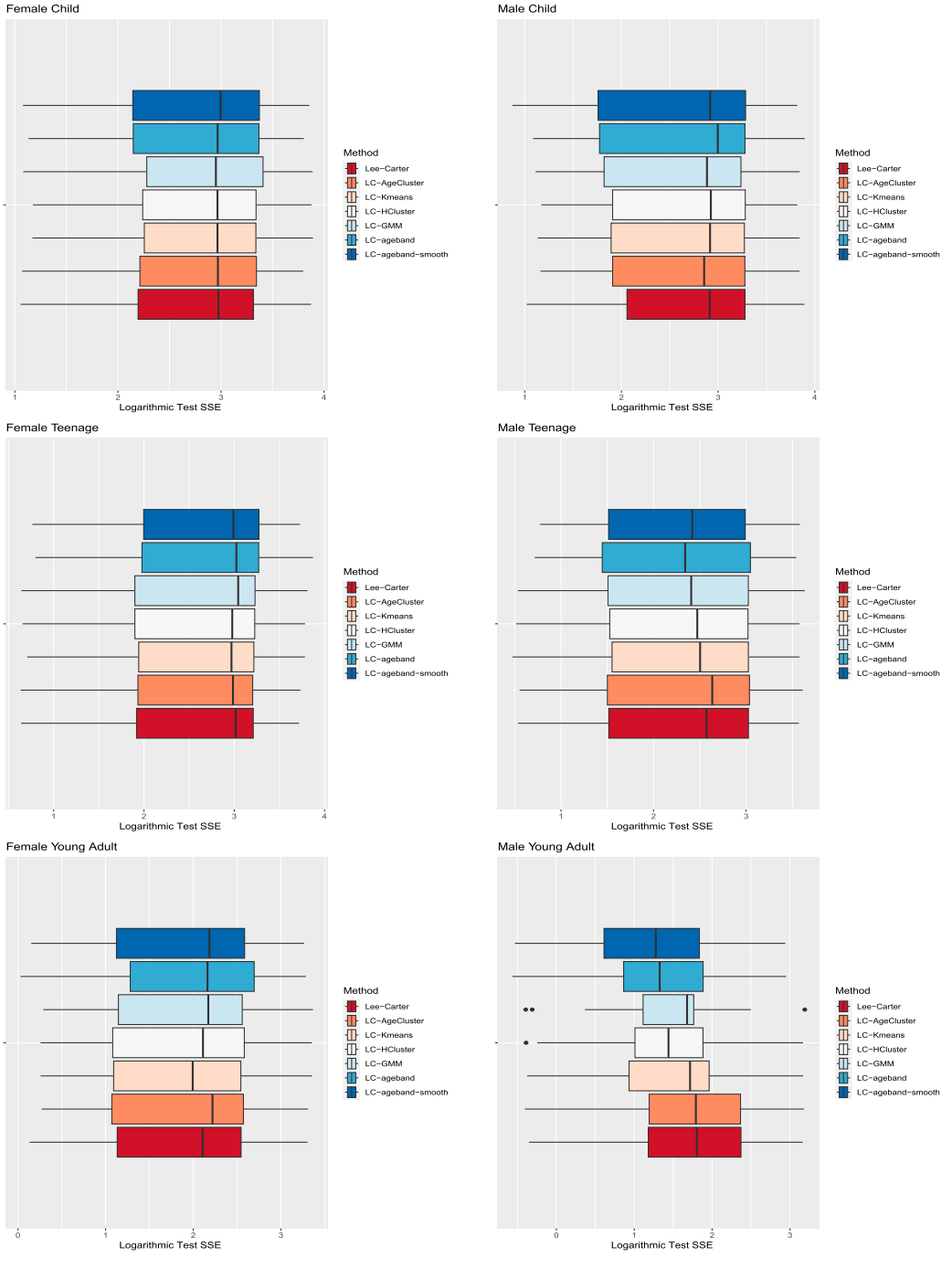


Figure 2. Boxplots of overall logarithmic test SSEs of 24 female populations (left panel) and 24 male populations (right panel) comparing the prediction performance of different methods for different age groups: Child, Teenage, and Young Adult (from the top to the bottom).

3.3. *Influence of smoothing*

The results in Section 3.2 have demonstrated an encouraging improvement in the overall prediction accuracy achieved by smoothing the radii of obtained age bands for individual target ages with the

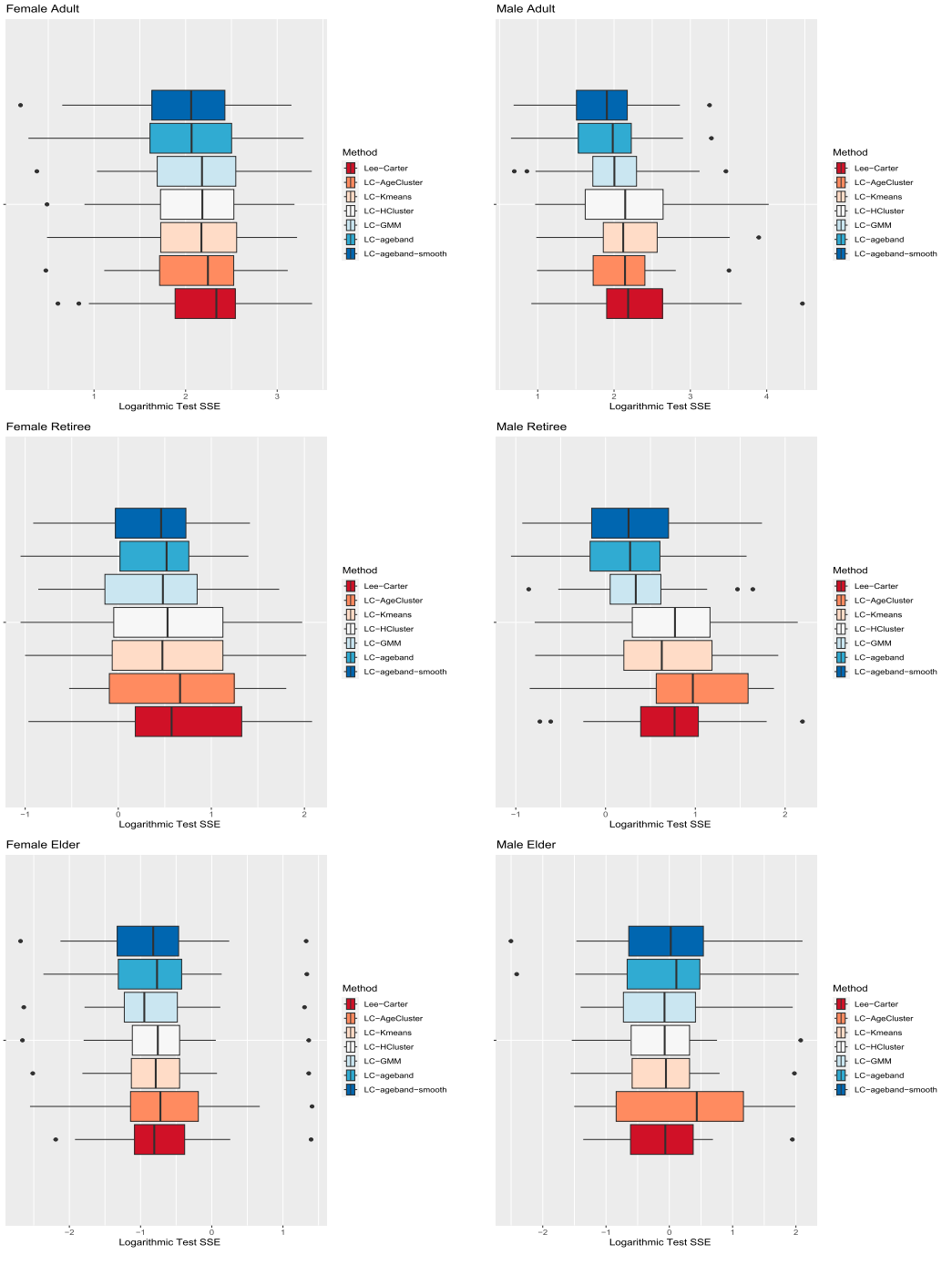


Figure 3. Boxplots of overall logarithmic test SSEs of 24 female populations (left panel) and 24 male populations (right panel) comparing the prediction performance of different methods for different age groups: Adult, Retiree, and Elder (from the top to the bottom).

smoothing window being $h = 3$ (compare rows 'LC-ageband' and 'LC-ageband-smooth' in Tables 2 and 3). In this section, we come to a closer look at how smoothing influences the effectiveness of the proposed framework.

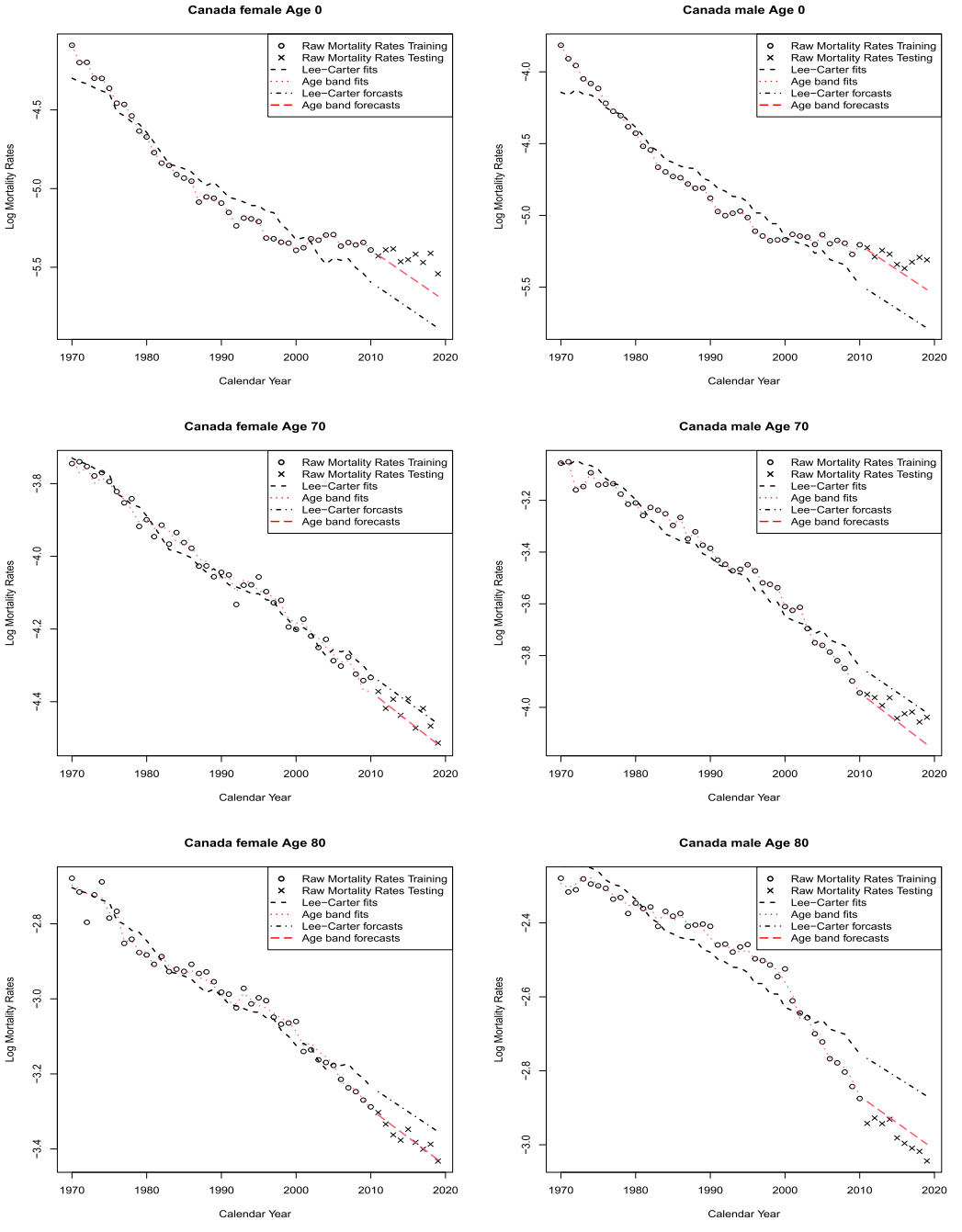


Figure 4. Fitted and the forecasted logarithmic mortality rates from the Lee-Carter model and the ASB method for specific ages chosen from Canadian data. Top to bottom: ages 0, 70, 80. Left to right: female, male.

We begin with illustrative plots of the functions r_x^* for several representative countries to develop intuition about how the function varies with age x. To strike a balance between filtering noise and preserving information, we plot the realized r_x^* sequence for each population and apply both the moving average procedure (with smoothing window $h = 3$) and a smoothing spline procedure to

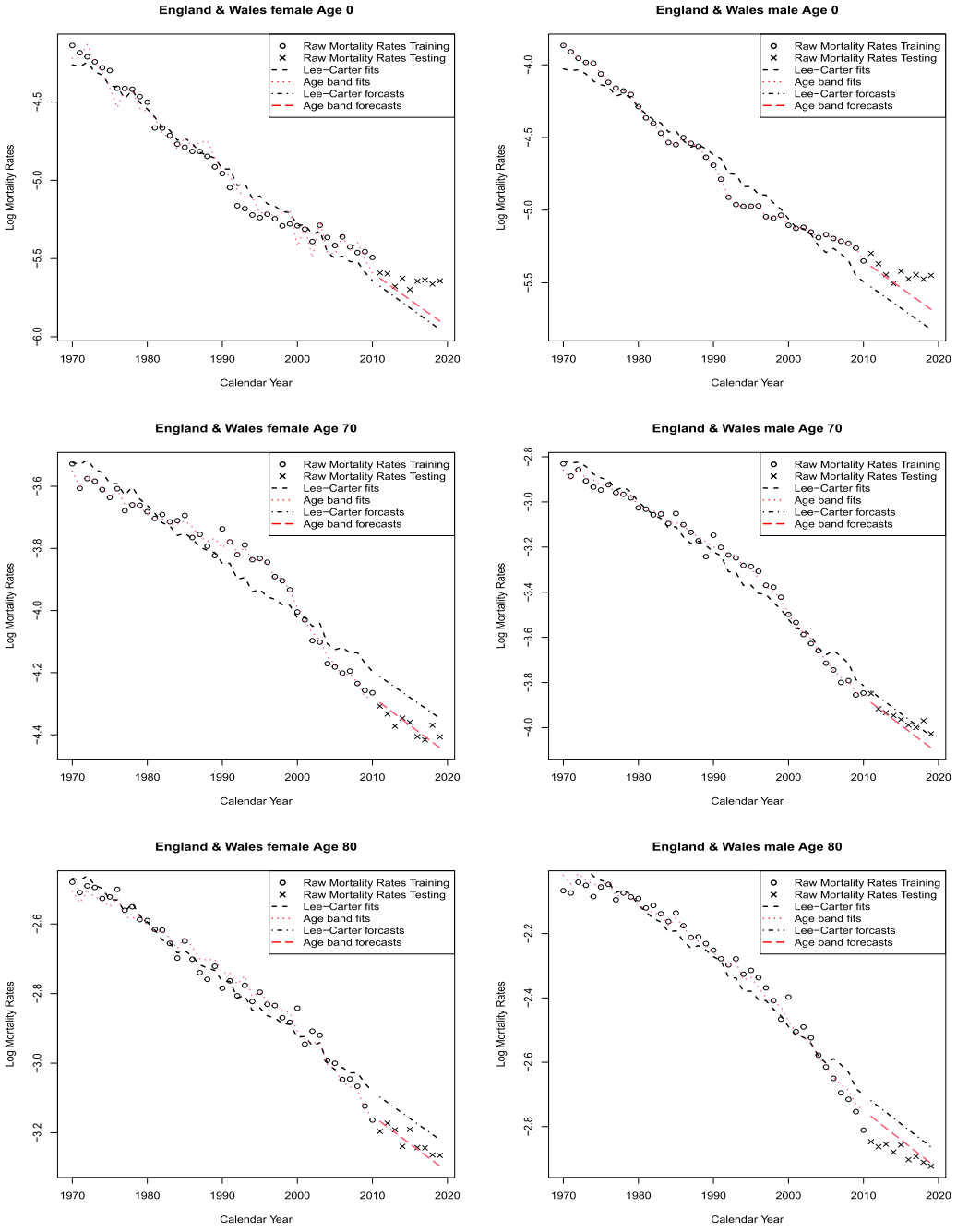


Figure 5. Fitted and the forecasted logarithmic mortality rates from the Lee-Carter model and the ASB method for specific ages chosen from England & Wales data. Top to bottom: ages 70, 80. Left to right: female, male.

the r_x^* sequence. The smoothing parameter in the spline procedure is determined by a conventional cross-validation step.

Figure 6 displays the age-specific band radius r_x^* for Canada, England & Wales, and France, for both genders. While the scale and number of peaks and troughs vary among different populations, the similarities among the results of both smoothing procedures have led to a general trend: except

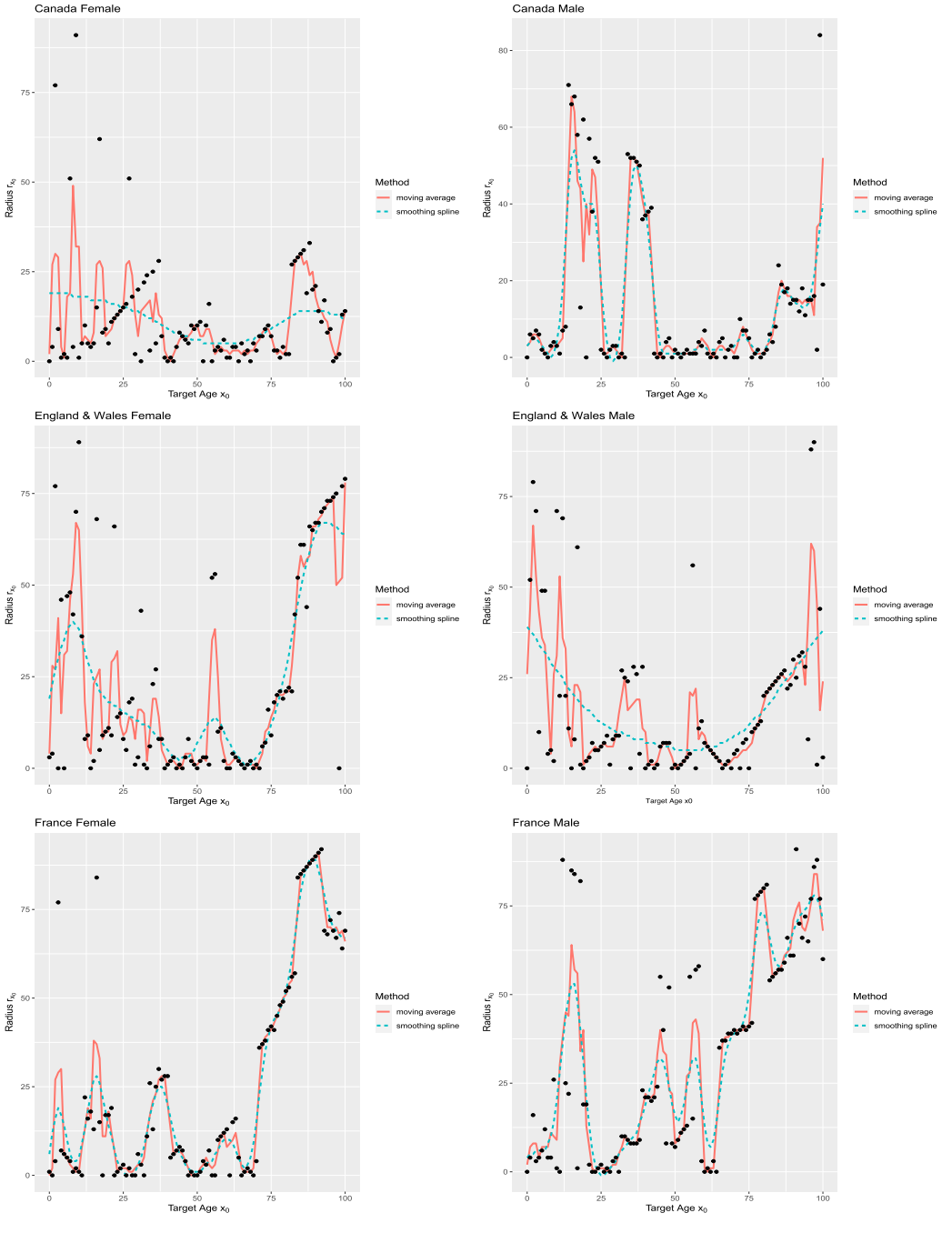


Figure 6. Age-specifc radius $r^*_{x_0}$ for some populations: black points represent the values of $r^*_{x_0}$ before a smoothing procedure; red solid lines indicate smoothed values from moving average with $h = 3$; blue lines represent the smoothed values from spline. Top to bottom: Canada, England & Wales, France. Left to right: Female, Male.

those ages close to infants, the radius tends to be larger at both lower and higher target ages compared to those middle values.

To examine whether different smoothing techniques for determining the band radius affect forecasting accuracy, we compared the resulting SSEs from three different smoothing procedures as

Table 4. Summary statistics of test SSEs of 24 female populations and 24 male populations comparing the prediction performance of the ASB methods under three different smoothing procedures: moving average with uniform $h = 3$ across all 24 populations in LC-ageband-smooth, moving average with population-specific h in LC-ageband-smooth-diff, and spline in LC-ageband-spline.

	1st Quartile	Median	Mean	3rd Quartile
	Female Population			
LC-ageband-smooth	23.92	58.37	57.71	84.61
LC-ageband-smooth-diff	22.79	58.09	57.82	85.35
LC-ageband-spline	**22.48**	**57.31**	**57.22**	**83.86**
	Male Population			
LC-ageband-smooth	21.38	**52.62**	48.14	72.13
LC-ageband-smooth-diff	22.00	54.27	**48.05**	**69.27**
LC-ageband-spline	**20.70**	53.38	48.99	71.47

Notes: Bold numbers highlight the minimum values in each column.

reported in Table 4. Below are descriptions of the smoothing procedures in each of the three predictive models reported in the table:

- **LC-ageband-smooth:** The moving average is applied with a uniform $h = 3$ across all the populations. The parameter value $h = 3$ is determined by the BCV procedure as explained previously, and the summary of the resulting SSEs has been presented in Table 2.
- **LC-ageband-smooth-diff:** The moving average is applied with a population-specific h derived by the BCV procedure for each population.
- **LC-ageband-spline:** Spline smoothing is applied for the determination of age band radius for each population.

A comparison of the SSEs among the three models in Table 4 indicates that all three smoothing procedures yield rather comparable overall SSEs across the 24 populations, suggesting no meaningful difference in forecasting accuracy across the three smoothing procedures.

In the rest of the section, we focus on the moving average procedure and examine how varying degrees of smoothness, represented by different values of h, affect the improvement in prediction performance. While the current choice of $h = 3$, determined through the BCV procedure by assuming a uniform smoothness across all 24 populations, yield improved predictive accuracy, understanding the model's sensitivity to this hyperparameter can provide insights into the potential robustness of the proposed method.

Figure 7 shows how averaged test SSEs over 24 populations change with respect to different values of smoothing window radius h for both male and female populations. A wide range of smoothing window lengths is considered, corresponding to scenarios of no smoothness ($h = 0$), some smoothness ($h = 1, 2, 3, \ldots, 10$), and strong smoothness ($h = 15$, 20, and 30). As indicated in the figure, the ASB method exhibits an obvious improvement over the Lee-Carter model. Skipping smoothing is apparently undesirable, and aggressive smoothing can be damaging, as increased test SSEs are observed for male population when $h = 15$, 20, and 30. The average test SSEs reach their optimal at $h = 2$ for males and $h = 10$ for females. However, we observe that the average test SSEs remains relatively stable for both male and female populations when the smoothing parameter h takes values from $h = 1$ to $h = 10$. This suggests that the prediction performance is not very sensitive to the choice of h as long as the value falls within a reasonable range.

One interesting question is whether there is a discernible pattern in the selected band radius as the target age varies from 0 to 100. To address this question, we calculate the average of the smoothed optimal band radius $r^*_{x_0,\mathbf{smooth}}$ over 24 populations and illustrate the values with respect to the target ages in Figure 8. The figure clearly demonstrates that smaller band radii are preferred for middle ages, whereas larger band radii are favored for younger and older ages.

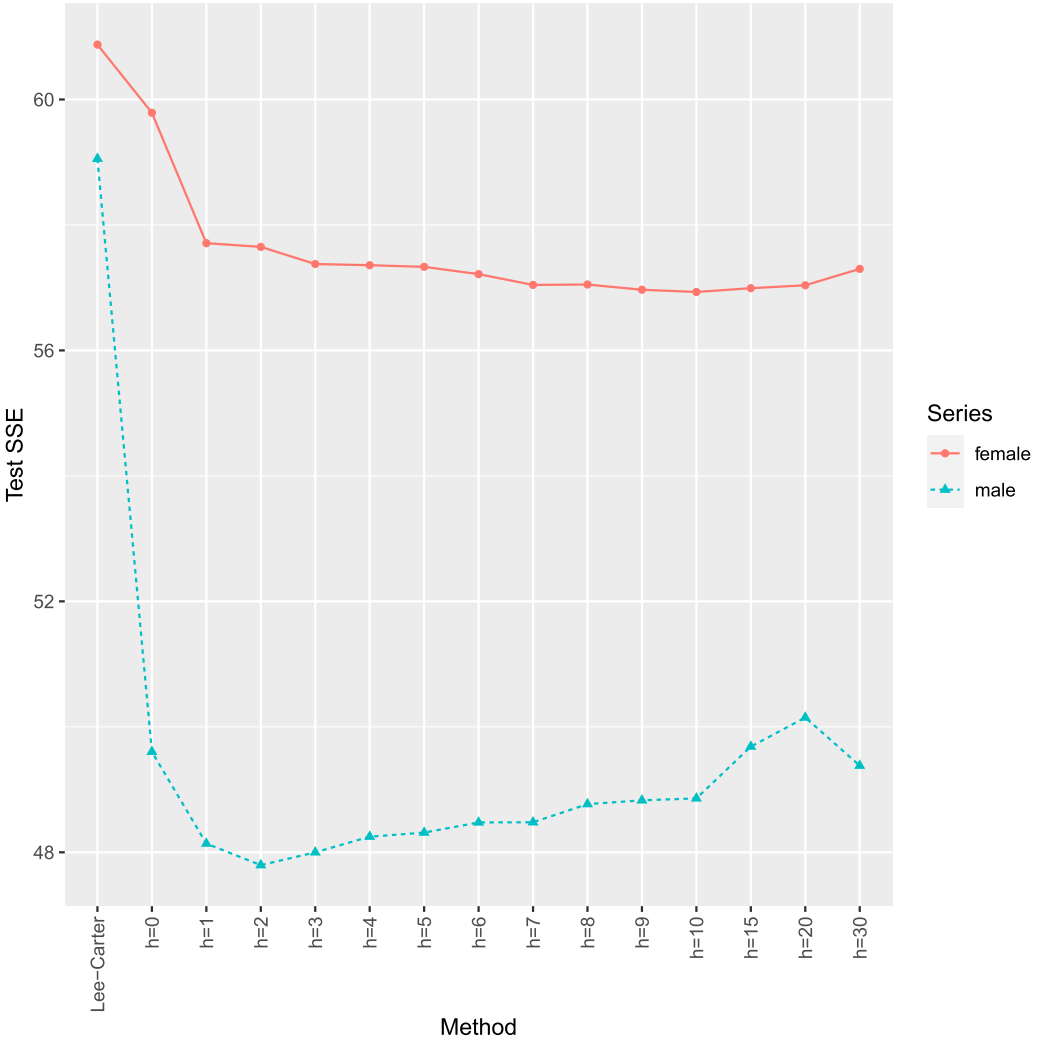


Figure 7. The Averaged values of test SSEs of 24 populations versus smoothing window radius *h*.

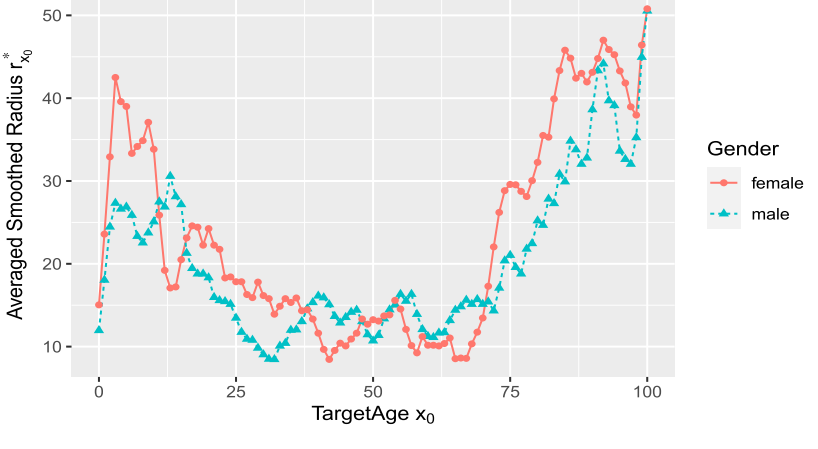


Figure 8. The averaged values of smoothed optimal band radius $r^*_{x_0,\mathbf{smooth}}$ versus target ages.



Figure 9. The averaged number of ages in selected bands versus target ages.

The radius can not fully reflect the number of ages included in the selected band for a target age, particularly for ages near the boundary ages 0 and 100. Therefore, we also calculate the average number of ages in the selected band after smoothing the radii for each target age over the 24 populations. The average number of ages in the selected band for each target age is illustrated in Figure 9. The pattern shown in the figure suggests that middle ages benefit from borrowing information from a moderate number of neighboring ages to improve their mortality prediction accuracy, while younger and older ages typically require more information and thus tend to borrow from a broader range of ages. However, there are exceptions for target ages of 0 or 1 (infants), where the average desirable band size tends to be smaller. This could be due to the unique mortality patterns among infants, which do not benefit significantly from including a large number of reference ages; instead, they typically rely on very few ages for modeling.

4. Age-specific set methods for multi-population data

In this section, we aim to make predictions for multiple populations by combining information from various mortality tables. We extend the foundational concept of the ASB method introduced in Section 2 to accommodate multiple-population data. To achieve this, we present two methods specifically designed to leverage information across ages and populations, enhancing mortality forecasts for each age.

4.1. Distance-based approach

We adopt the age-specific set paradigm introduced in Section 2.1 to decompose the overall prediction objective for a target population's mortality rates into micro-objectives of predicting each individual age based on data from multiple populations. In this expanded context, the information pool includes the mortality development trajectories of all ages of all populations. The proposed method must effectively extract useful information while filtering out noise to enhance prediction accuracy for the target age, all while maintaining computational efficiency. To achieve this, we generalize the concept of the age-specific band. In the single-population context, the age-specific band comprises a set of consecutive neighboring ages, as described in Section 2.2. However, in a multiple-population context, the notion of 'neighborhood' for a target age shifts from its intuitive understanding based on natural age ordering to identifying those ages that share the most similarities with the target age across the entire information pool. To operationalize this approach, we propose a variance-based metric to quantify the distance between two mortality ages and define the 'neighborhood' based on

the distance measure, thereby expanding the age-specific set paradigm to enable information borrowing from multiple populations. We generally call the mortality forecasting method based on such selection of the 'neighborhood' as the distance-based approach.

Two parallel mortality trajectories at different levels are considered to share the same developmental trend in mortality. Consequently, the proposed distance measure is designed to overlook differences caused by varying mortality levels and to focus instead on changes over time. The distance measure between two mortality trajectories is established through the following two steps:

(1) Define the difference sequences as

$$\text{diff}_{x,i;y,j}(t) = \eta_{x,i}(t) - \eta_{y,j}(t), \quad t \in \mathcal{T},$$

where $\eta_{x,i}(t)$, $t \in \mathcal{T}$, represents a mortality trajectory for age x in the ith population and the choice of $\eta_{x,i}$ depends on the chosen mortality model. $\eta_{x,i}(t)$ can take the form of $\log m(x,i,t)$ for Lee-Carter model and $\text{logit}\, q(x,i,t)$ for CBD model. '$\text{diff}_{x,i;y,j}(t)$' represents the difference between the mortality sequence of age x in population i and that of age y in population j.

(2) Define the distance between two mortality trajectories as the variance of the difference sequence:

$$D_{x,i;y,j} = \text{Var}\left(\text{diff}_{x,i;y,j}(t)\right).$$

By definition, $D_{x,i;y,j}$ is a dissimilarity measure. A lower value of $D_{x,i;y,j}$ indicates a greater resemblance of the two mortality sequences and a closer alignment in their mortality development patterns over time. $D_{x,i;y,j} = 0$ for two parallel sequences which have a constant difference over time.

The proposed distance-based approach consists of the following steps:

(1) For a target age x_0 of a given population i_0, the distance measure $D_{x_0,i_0;y,j}$ is calculated for every y across the entire age range and every population j. The resulting sequences are then sorted in ascending order based on their distances from the target age in the target population.

(2) A constant, denoted as K, which represents the maximum allowed number of sequences to borrow information from, is predetermined. This is a hyperparameter in the proposed distance-based approach. A large number of K permits more models for choice in the modeling procedure, while it also yields heavier computational demands.

(3) Among the set of K nearest 'neighbors', $k(\leq K)$ neighboring ages are selected for a joint modeling with the target age. The parameter k can take possible values in $\{0, 1, \ldots, K\}$ and it determines the size of the 'neighborhood' of the target age chosen for joint modeling. The determination of k follows a 4-fold BCV procedure, similar to the one used for the selection of the optimal radius $r^*_{x_0}$ in the age-specific band method detailed in Section 2.2. We denote the value of k from the 4-fold BCV by $k^*_{(x_0,i_0)}$ with the subscripts to signify that it is specific to age x_0 of population i_0.

(4) Similar to the argument in Section 2.2.3, a smoothing step is recommended to mitigate the potentially abrupt fluctuations in the value of $k^*_{(x_0,i_0)}$ with x_0 varying over the full age range. In the empirical studies in Section 4.3, we employ the same moving average smoothing scheme as described in Section 2.2.3 with a smoothing window radius $h = 3$ for both male and female populations. The resulting smoothed 'neighborhood' size is demoted as $k^*_{(x_0,i_0).\textbf{smooth}}$ and the final age-and-population specific set $\mathcal{A}^*_{(x_0,i_0)}$ determined for joint modeling with age x_0 of population i_0 consists of the $k^*_{(x_0,i_0).\textbf{smooth}}$ ages in its neighborhood defined by the distance measure $D_{x,i;y,j}$.

(5) The respective mortality sequences associated with the ages in $\mathcal{A}^*_{(x_0,i_0)}$ are used to calibrate the underlying mortality model and make predictions for age x_0 of population i_0.

The predetermined value of the maximum permissible number of ages, K, can impact the performance of the proposed distance-based approach. A larger K value provides a greater potential for improving predictions by leveraging a broader candidate pool but introduces a heavier computational burden. Conversely, a smaller K value assumes that the most valuable information is confined within ages with strong similarities only. In the empirical studies discussed in Section 4.3, we explore the options of $K = 100$, 200, and 400 to study the marginal improvements in prediction accuracy achieved by increasing the value of K.

4.2. ACF model-based approach

A factor worth considering for further improving the effectiveness of borrowing information across ages and populations is the choice of the underlying mortality model. In the preceding subsection, Lee-Carter model is embedded in the distance-based approach. In this subsection, we consider an approach embedded with the ACF model (N. Li & Lee, 2005), a multiple-population mortality model.

Let $m_{x,t,i}$ denote the central death rate or age-specific death rate (ASDR) of the ith population in a group G of populations for age x and year t. The ACF model depicts the mortality development of each individual population in the following form:

$$\log m_{x,t,i} = a_{x,i} + B_x K_t + b_{x,i} k_{t,i} + \epsilon_{x,t,i}, \quad t = t_L + 1, \ldots, t_U, \tag{9}$$

with normalization constraints $\sum_x B_x = 1$, $\sum_t K_t = 0$, $\sum_x b_{x,i} = 1$, and $\sum_t k_{t,i} = 0$, where $\epsilon_{x,t,i}$'s are the i.i.d. error terms. The static age functions $a_{x,i}$ are still estimated as the average of logarithmic ASDRs over the modeling time period for each population:

$$\hat{a}_{x,i} = \frac{1}{t_U - t_L} \sum_{t=t_L+1}^{t_U} \log m_{x,t,i}, \quad i \in G. \tag{10}$$

For the estimation of the other parameters, the SVD procedure is applied to the aggregate data of the group to obtain B_x and K_t, and then an extra round of SVD is further applied to the residual data, $(\log m_{x,t,i} - \hat{a}_{x,i} - \hat{B}_x \hat{K}_t)$, to get $b_{x,i}$ and $k_{t,i}$ for each individual population.

The age set selection comes with various options. Instead of making a selection from a complete pool containing all ages and populations, we consider selecting the age set and population separately based on different criteria via separate procedures. As for population selection, we divide populations into geographic regions based on exogenous geographic information and use the ones in the same geographic region for predicting the mortality of a target population. The geographic division of populations has been a standard practice in multi-population mortality modeling literature; see, for instance, Richman and Wuthrich (2018). Populations in the same geographical region are considered to have more similar mortality trends and thus are more appropriately modeled jointly by a multiple-population model, for instance, the ACF model.

For the age set selection, we adopt the age band selection paradigm introduced in Section 2. More specifically, we consider a sequence of age bands of different radii for each target age x_0 from a specific target population and employ the BCV procedure to determine the optimal band radius. A smoothing step of the moving average scheme with $h = 3$ is then applied to obtain the smoothed age-specific band radii. After selecting the population and age band, an age-specific ACF model is built for every target age of the target population, and the prediction of mortality is then made based on a projection using the resulting ACF model.

4.3. Empirical studies with multi-population mortality data

In this section, we empirically evaluate the performance of the distance-based and ACF model-based approaches introduced in the preceding two subsections using the same mortality data from the HMD

Table 5. Summary statistics of test SSEs from the 24 population comparing the prediction performance between distance-based models (borrowing information across populations), and ASB models (borrowing information within single population).

	1st Quartile	Median	Mean	3rd Quartile
Female Population				
LC-ageband	22.79	61.14	59.79	86.51
LC-ageband-smooth	23.92	58.37	57.71	84.61
multi-LC-distance ($K = 100$)	24.30	61.85	60.54	91.13
multi-LC-distance ($K = 200$)	24.43	61.10	59.50	87.12
multi-LC-distance ($K = 400$)	24.03	62.41	59.40	86.93
multi-LC-distance-smooth ($K = 100$)	22.12	56.75	56.58	**81.78**
multi-LC-distance-smooth ($K = 200$)	22.10	56.27	55.95	82.35
multi-LC-distance-smooth ($K = 400$)	**21.90**	**55.82**	**55.90**	83.57
Male Population				
LC-ageband	**20.39**	55.76	49.60	71.87
LC-ageband-smooth	21.38	52.62	48.14	72.13
multi-LC-distance ($K = 100$)	21.37	54.51	48.27	68.62
multi-LC-distance ($K = 200$)	21.33	54.03	47.95	67.59
multi-LC-distance ($K = 400$)	21.34	53.75	47.73	67.53
multi-LC-distance-smooth ($K = 100$)	21.27	52.47	45.37	63.48
multi-LC-distance-smooth ($K = 200$)	21.72	52.56	45.10	**61.91**
multi-LC-distance-smooth ($K = 400$)	21.73	**51.80**	**45.05**	62.67

Notes: Bold numbers highlight the minimum values in each column.

analyzed in Sections 3. We apply the same partition of the data into a training set and a test set as in Sections 3.

We explore two distance-based models under multi-population settings, which borrow information across all 24 populations and thus form a pool containing ages 0–100 from each of these 24 populations, totaling 2424($= 101 \times 24$) candidates.

- **multi-LC-distance**: This method is implemented as described in Section 4.1, ranking the distances of 2423($= 2424 - 1$) candidate ages to the target age x_0. The term 'multi' in the name of the method indicates that we allow borrowing information across populations, while 'LC' refers to the underlying mortality model which is the Lee-Carter model.
- **multi-LC-distance-smooth**: A smoothing procedure with $h = 3$ is applied to the optimal 'neighboring' sizes obtained from **multi-LC-distance**.

These two methods are implemented with $K = 100$, 200 and 400 to study the increase in prediction accuracy with respect to an increased maximum information borrowing capacity. The overall prediction accuracy is evaluated by the overall test SSEs as in Section 3. Table 5 reports summary statistics of test SSEs from the 24 populations for the distance-based approach mentioned in this section, along with the results from the Lee-Carter model, LC-ageband, and LC-ageband-smooth for comparison.

From Table 5, we have the following observations:

(1) Notably, applying the smoothing procedure significantly improves prediction accuracy for the distance-based approaches, as evidenced by comparing the 'multi-LC-distance' rows with the 'multi-LC-distance-smooth' rows for both female and male populations.
(2) For both female and male populations and whenever smoothing is applied or not, the mean of the resulting 24 test SSEs exhibits a decreasing trend as K increases. In other words, enhanced predictive performance can be attained by expanding the candidate pool through an increase in K. However, it is important to acknowledge that the implementation of the multi-population distance-based method becomes computationally demanding as the value of K increases while the marginal gains in predicting accuracy decrease.

Table 6. Summary statistics of test SSEs of the 24 populations comparing the prediction performance between the distance-based and ACF model-based methods.

	1st Quartile	Median	Mean	3rd Quartile
Female Population				
multi-LC-distance-smooth ($K = 400$)	**21.67**	**55.98**	**55.80**	84.03
ageband-ACFGeoInfo	23.98	58.79	56.02	**83.27**
Male Population				
multi-LC-distance-smooth ($K = 400$)	22.65	51.84	45.26	61.90
ageband-ACFGeoInfo	**21.03**	**46.24**	**42.77**	**59.45**

Notes: Bold numbers highlight the minimum values in each column.

(3) For both male and female populations, the multi-population approaches lead to noticeable improvements in prediction accuracy compared to the single-population approaches.

In the rest of the section, we compare the performance of the distance-based approach with the ACF model based method. In Table 6, we report the results of the best performed distance-based approach 'multi-LC-distance-smooth, $K = 400$' from Table 5 and add results of ACF model based approach. For ACF model based method, population selection is based on a geographical division given in Table 1 and age set selection is based on age-population specific radius for each target age x_0 from a specific target population via the block cross-validation procedure. A smoothing step of the moving average scheme with $h = 3$ smoothed with a smoothing window radius $h = 3$ for both male and female populations. The ACF model-based method (i.e. 'ageband-ACFGeoInfo' in Table 6) shows a notable improvement in predicting accuracy over the distance-based method for male populations, but performs slightly worse for female populations.

5. Concluding remarks

Observations on similarities in mortality development trajectories of neighboring ages are widely made in mortality data and offer opportunities for enhancing mortality prediction accuracy. Conversely, dissimilarities across distinct age ranges, such as adult versus senior ages, are also noticeable. Conventional approaches typically model all ages together using a generic model, which may not always be the optimal strategy. When the focus is on improving predictive accuracy, overlooking the potential drawbacks of using data with an inappropriate scope and incorporating data with conflicting signals can actually weaken the model's performance. By carefully selecting the most relevant data, it is possible to maintain the model's simplicity while also boosting its effectiveness. Our contribution lies in proposing a framework that customizes the selection of data range for each specific target age, drawing information from either a single population or multiple populations. This introduces a fresh perspective on mortality forecasting and highlights the importance of incorporating age-specific data into extrapolative models to enhance forecasting accuracy. Moreover, the proposed framework is versatile to embed many existing mortality models. While the empirical studies in this paper focus on the utilization of the Lee-Carter model and the ACF model, the framework can accommodate many other mortality models.

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Appendix. The one-sided Diebold-Mariano (DM) test

In this appendix, we describe the general procedure of the Diebold-Mariano (DM) test. Let M_1 and M_2 denote the two models under comparison. Below is a step-by-step description of the procedure:

(1) Calculate forecast errors defined as $e_{i,t} = y_t - \hat{y}_{i,t}, t = 1, \ldots, T$, for each model $M_i, i = 1, 2$, where y_t is the actual value and $\hat{y}_{i,t}$ is the forecast for period t from model i, and T is the total number of forecast periods.
(2) Calculate loss differentials $d_t = L(e_{1,t}) - L(e_{2,t}), t = 1, \ldots, T$, where L is a loss function, commonly taken as the squared error loss $L(e) = e^2$.
(3) Calculate the DM test statistic given by:

$$DM = \frac{\bar{d}}{s_d/\sqrt{T}}$$

where $\bar{d}$ is the mean of the loss differentials calculated by

$$\bar{d} = \frac{1}{T}\sum_{t=1}^{T} d_t$$

and s_d^2 is the sample variance of the loss differentials calculated by

$$s_d^2 = \frac{1}{T-1}\sum_{t=1}^{T}(d_t - \bar{d})^2$$

(4) Impose the following one-side test hypotheses:

$$H_0 : E[d_t] \leq 0 \quad \text{versus } H_a : E(d_t) > 0.$$

Assume that the loss differentials $\{d_1, \ldots, d_T\}$ is stationary in the sense that they have the same mean and variance. Then, under the null hypothesis, the test statistic DM asymptotically follows a standard normal distribution as the sample size $T \to \infty$.
(5) Calculate p-value$= 1 - \Phi(DM)$ where Φ denotes the standard normal distribution. For a given significance level α (e.g.0.05), we reject the null hypothesis H_0 if p-value$< \alpha$, and accept H_0 otherwise.